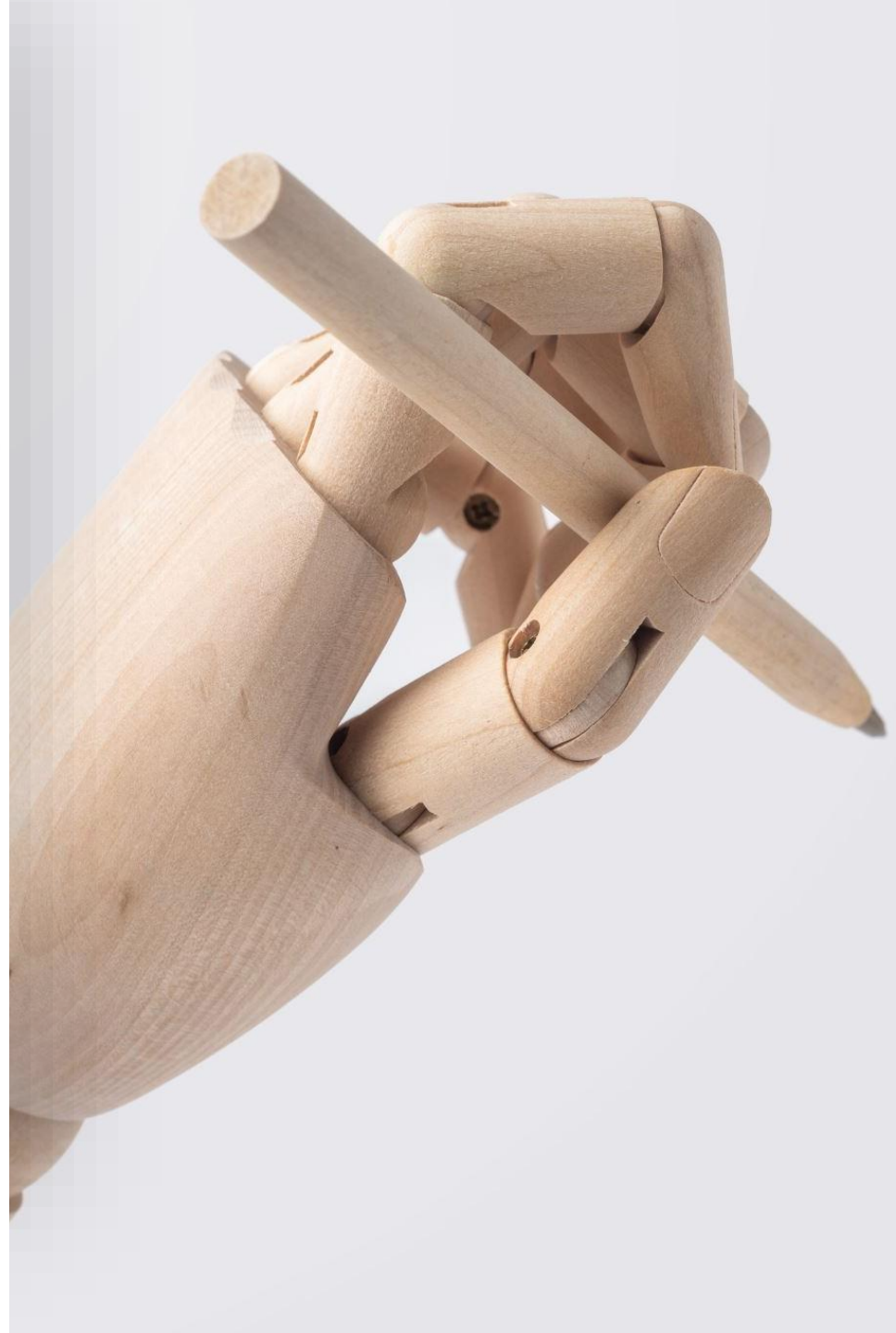


Evaluation of the Elbow and Wrist

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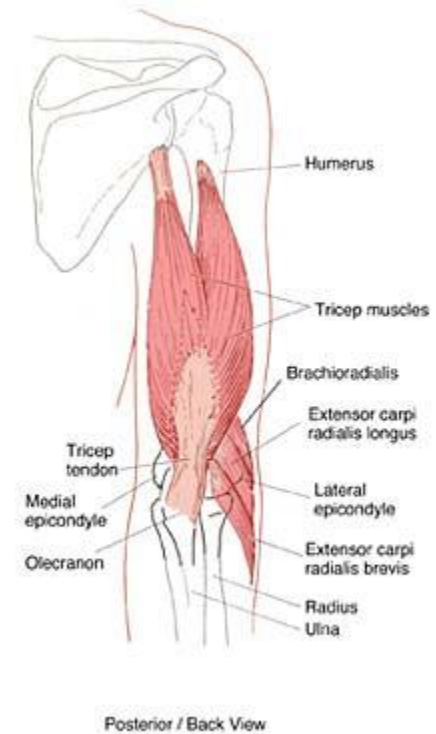
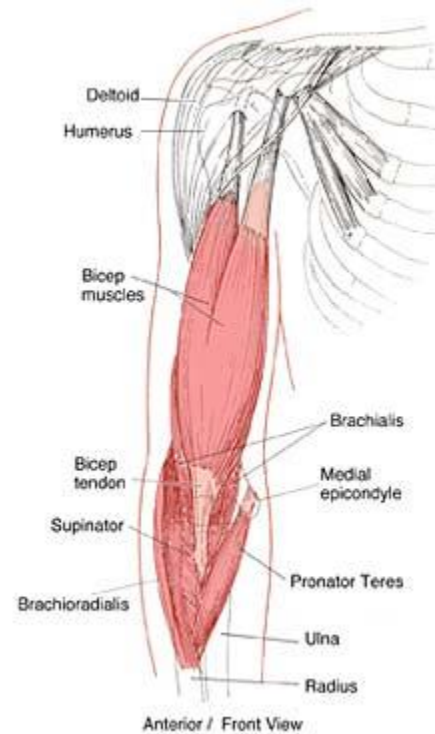


Overview

- Common in athletes, laborers, and desk workers
- Usually overuse, trauma, or nerve entrapment

Elbow

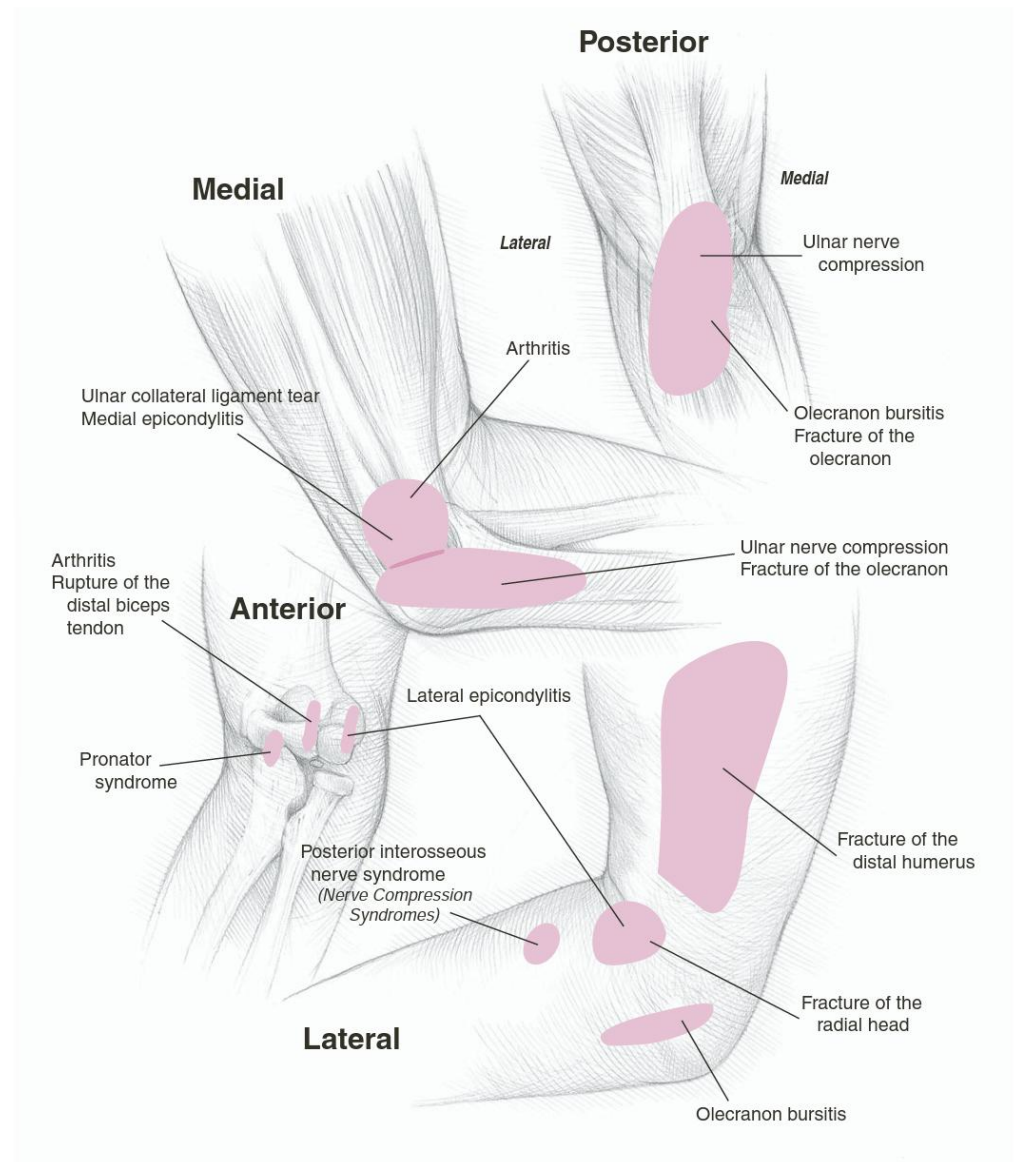
- Soft Tissue
- Orthopedic Structures
- Neurologic Structures



Common Causes

PAIN DIAGRAM Elbow and Forearm

- Lateral – Tennis elbow (wrist extensor strain)
- Medial – Golfer's elbow (wrist flexor strain)
- Posterior – Olecranon bursitis
- Anterior – Biceps tendon rupture, pronator syndrome
- Referred pain
 - Cervical radiculopathy
 - Shoulder pathology
 - Rotator cuff tendinopathy



Focused History

- Onset: overuse vs trauma
- Pain location – one-finger rule
- Activities – gripping, lifting, throwing
- Numbness/tingling – ulnar or radial distribution
- Red flags: swelling, erythema, systemic signs

Elbow

- Inspection
- Palpation
 - Neck
 - Shoulder
 - Elbow
- Range of Motion
 - Neck
 - Shoulder
 - Elbow
- Neurovascular Exam
- Strength- joint above/below (Shoulders, Elbow, hand/wrist)
- Special Tests
 - Resisted Supination
 - Resisted Pronation
 - Cozen
 - Tinels Test of Elbow
- OPP considerations
 - Radial head
 - Interosseous membrane

Inspection

- Compare both elbows for symmetry, deformity
- Inspect the elbows for swelling, erythema or ecchymosis
- Assess extremities for muscle atrophy or wasting and any skin lesions



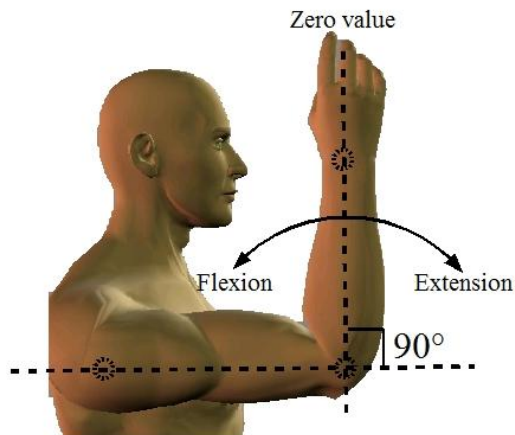
Palpation

- Palpate the olecranon, proximal radial head to assess for fracture or dislocation
- Palpate the medial and lateral epicondyles to assess for pain or tenderness



Active Range of Motion

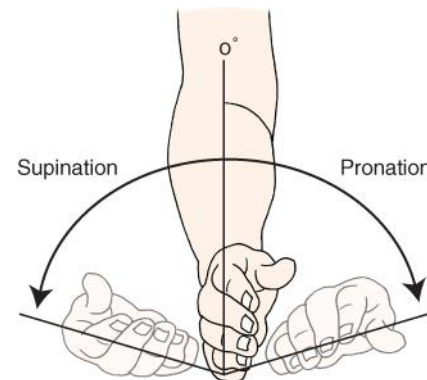
- Flexion
- Extension
- Pronation
- Supination
- If AROM restricted, perform PASSIVE ROM



Passive Range of Motion

Hold the patient's wrist and perform:

- Flexion
- Extension
- Pronation
- Supination



Strength & Neuro

Strength: resisted wrist flex/extension, elbow flex/extension

Weakness ? Consider sensation and reflexes.

Neuro: ulnar (T1), median (C6–7), radial (C7–8)

Elbow Special Tests

Special tests:

-
- Cozen's – lateral epicondylitis
 - Reverse Cozen's – medial epicondylitis
 - Tinel's – cubital tunnel syndrome



Cozen's Test

(lateral epicondylitis)



- Positive if patient reports pain in the lateral epicondyle with resistance.

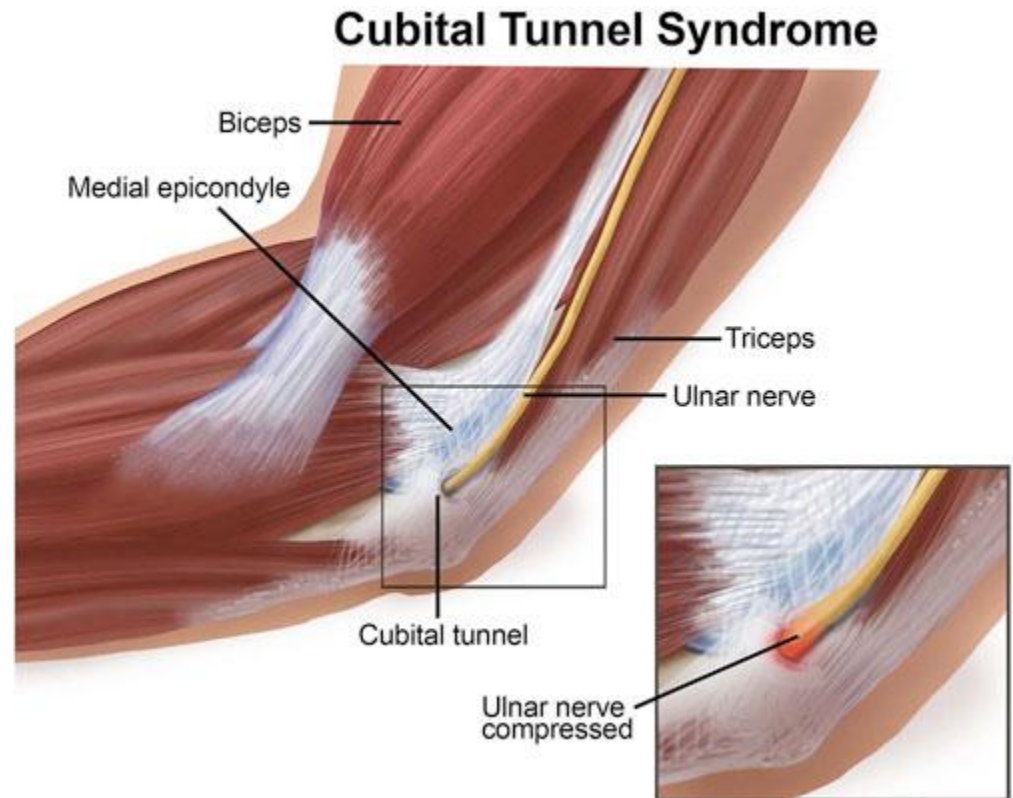
Reverse Cozen's test (medial epicondylitis)

- Positive if patient reports pain in the medial epicondyle with resistance.



Tinel's Test (elbow)

- Tests for Cubital Tunnel Syndrome
- A positive test replicates sensation of numbness/tingling in ring and pinky (Ulnar Nerve)
- Tinel's Test Elbow



Clinical Pearls

Pain location = diagnosis map

Lateral = wrist extensors

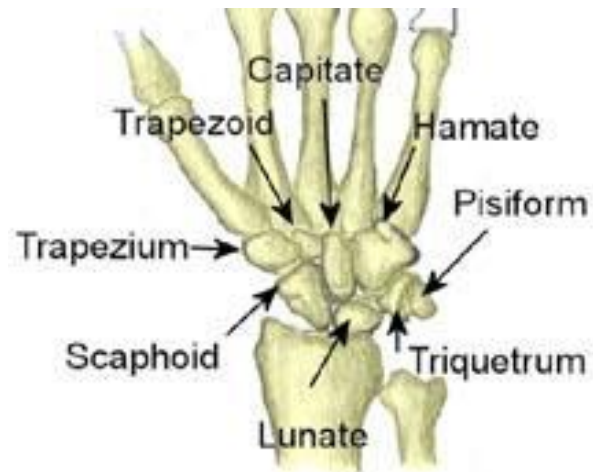
Medial = wrist flexors

Posterior = olecranon

Summary

- Elbow pain is often mechanical/overuse
- A systematic exam helps localize structure that is affected
- Red flags → infection or trauma assessment

Wrist



DDx for *wrist* complaint

- Traumatic
 - Sprain
 - Scaphoid fracture
 - Distal radial fracture
- Non-traumatic
 - Arthritis
 - Dorsal ganglion cyst
 - Carpal Tunnel syndrome
 - De Quervian's tenosynovitis

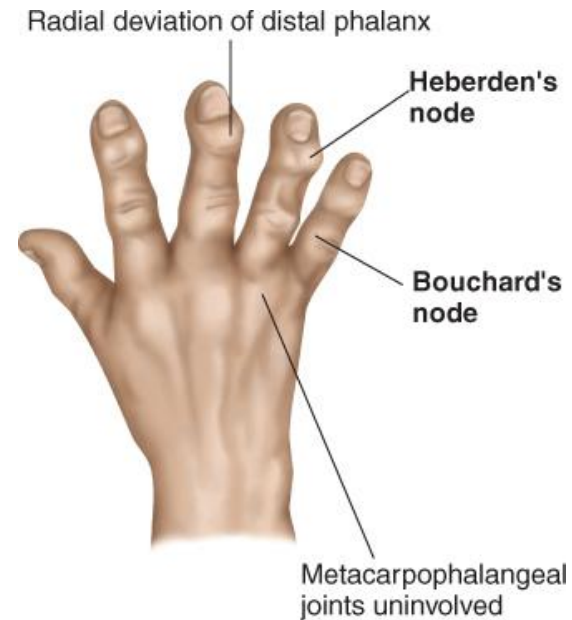
WRIST EXAM

Hand/Wrist

- Inspection
 - Palpation
 - Neck
 - Elbow
 - Range of Motion
 - Neck
 - Shoulder
 - Elbow
 - Neurovascular Exam
- Strength- joint above/below (elbow, grip strength)
 - Special Tests
 - Tinel's sign
 - Phalen's test
 - Finkelstein Test
 - OMM considerations
 - wrist bones
 - Fascia of the wrist

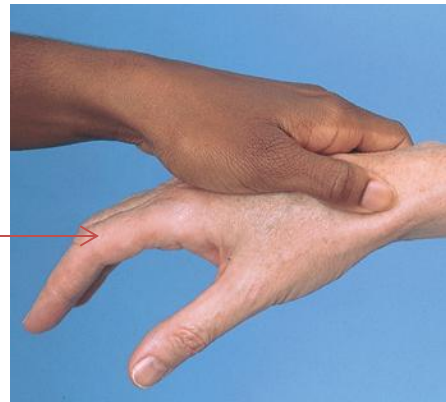
Inspection

- Compare both wrists and hands for symmetry
- Inspect wrists for any swelling or deformities
- Assess the palmar surface of the hands for atrophy of the thenar and hypothenar eminence



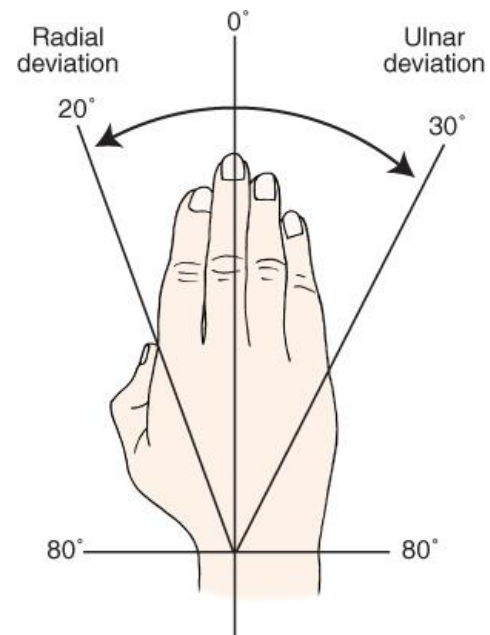
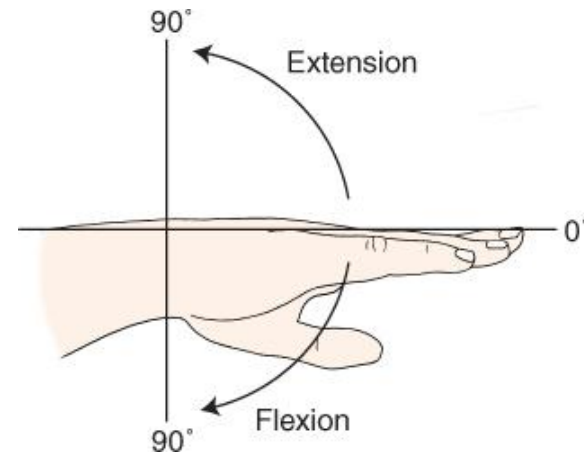
Palpation

- Palpate the distal radial head for fracture or dislocation
- Palpate the “anatomical snuffbox” to assess for scaphoid fracture
- Palpate the carpal and metacarpal bones for pain or tenderness
- Squeeze the metacarpal bones to assess for pain
- Palpate the DIP and PIP joints of each finger to assess for tenderness or deformity



Active Range of Motion

- Flexion
- Extension
- Ulnar deviation
- Radial deviation



Passive Range of Motion

Hold the patient's forearm proximal to the wrist and perform:

- Flexion
- Extension
- Ulnar deviation
- Radial deviation



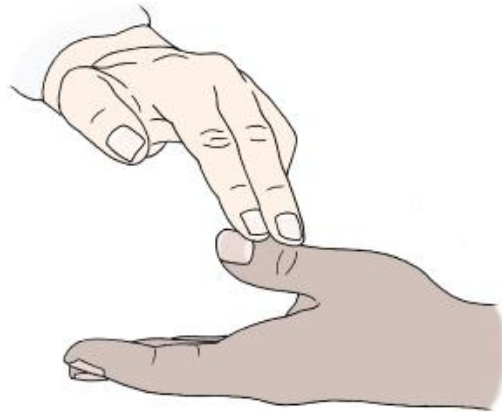
Grip Strength

- Have the patient grab your second and third fingers tightly while you try to pull your fingers away.

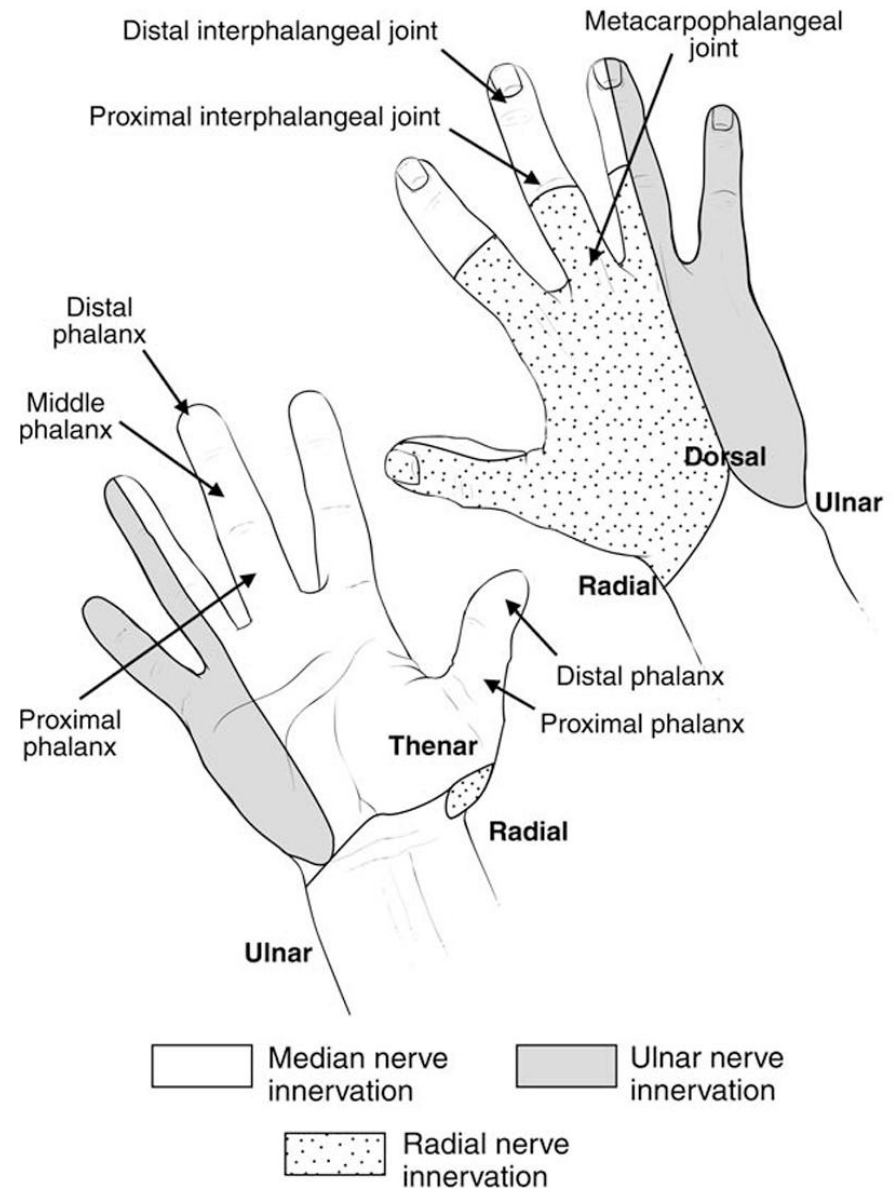


Strength Testing

- Test thumb abduction by having the patient raise their thumb straight up as you apply downward resistance. Weakness could suggest *Carpal Tunnel Syndrome*.



Nerve distribution



WRIST SPECIAL TESTS



Tinel's Sign at the wrist

- Positive test is pain or paresthesia's of the first three fingers consistent with *Carpal Tunnel Syndrome (Median nerve distribution)*
- [Tinel's Sign for CTS](#)



Phalen's Test

- A positive test is recreation of paresthesia's in first three fingers (*median nerve distribution*) suggesting *carpal tunnel syndrome*
- [Phalen's Test](#)



Carpal Compression Test

- Patient position in standing or sitting. The patient forearm is supinated and then the examiner applies direct pressure over the carpal tunnel (median nerve) between the thenar and hypothenar eminence for 30 seconds.
- A positive test is recreation of paresthesia's in first three fingers (*Carpal Tunnel Syndrome*).
- [Carpal Compression Test](#)



Finkelstein's Test

- Have the patient flex their thumb and clench fist over thumb. Push the wrist into ulnar deviation. Positive test is pain at the radial aspect of thumb and wrist, consistent with *DeQuervain's Tenosynovitis*.
- [Finkelstein's Test](#)





**"IT'S A TORN ROTATOR CUFF-- YOU CAN
FORGET ABOUT THROWING THUNDERBOLTS FOR AWHILE..."**